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Simon Pedro Galeano Muñoz

PhD student in Statistics

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PhD student in Statistics at the King Abdullah University of Science and Technology (KAUST), advised by Prof. Maurizio Filippone, working at the intersection of information theory, generative modeling, and statistical estimation. My research centers on score-based diffusion models and neural estimators of information-theoretic quantities such as mutual information and transfer entropy, with broader interests in Bayesian deep learning, stochastic processes, and scalable approximate inference.

PUBLICATIONS

Transfer Entropy Neural Diffusion Estimation (TENDE)

S. P. Galeano Muñoz, M. Bounoua, G. Franzese, P. Michiardi, M. Filippone.

AISTATS 2026

DIPHINE: Diffusion-based Φ -ID Neural Estimator

S. P. Galeano Muñoz, M. Bounoua, G. Franzese, P. Michiardi, M. Filippone.

Under review, 2026

EDUCATION

Ph.D. in Statistics, *King Abdullah University of Science and Technology (KAUST)*

Jan 2026 — Present

Advised by Prof. Maurizio Filippone.

M.Sc. in Statistics, *King Abdullah University of Science and Technology (KAUST)*

Aug 2024 — Dec 2025

B.Sc. in Statistics (Top graduate of the class), *Universidad Nacional de Colombia at Medellín*

Apr 2024

EXPERIENCE

Graduate Teaching Assistant

Jan 2025 — Present

King Abdullah University of Science and Technology

Thuwal, Makkah Province, Saudi Arabia

- Run tutorials for graduate students to reinforce their understanding of probability and applied statistics.

Advanced Analytics Intern

Aug 2023 — Jan 2024

TIGO Colombia

Medellín, Antioquia, Colombia

- Built machine learning models, deployed in production, to estimate the probability of a client returning a device, and optimized the data-preprocessing pipeline to automate model retraining.
- Generated visualization dashboards highlighting business KPIs for decision-making.

Research Intern

Jan 2023 — Jul 2023

King Abdullah University of Science and Technology

Thuwal, Makkah Province, Saudi Arabia

- Evaluated the performance of parameter estimation in geostatistical models.
- Carried out stability and sensitivity analysis of solutions to stochastic partial differential equations.

Teaching Assistant (Statistics II & Descriptive Statistics)

Sep 2021 — Dec 2022

Universidad Nacional de Colombia at Medellín

Medellín, Antioquia, Colombia

- Supported over 150 students with theoretical questions on regression models and exploratory data analysis.
- Designed supplementary course material, contributing to a low failure rate (under ten percent).
- Taught introductory R programming for fitting and interpreting regression models.

AWARDS AND ACHIEVEMENTS

- Placed 1st in the [Statistics Qualifying Examinations](#) at KAUST. **Dec 2024**
- Honorary degree (highest GPA among graduates of the Faculty of Science in the respective cohort). **Apr 2024**
- Winner of the [2023 KAUST Competition on Spatial Statistics for Large Datasets](#) (Sub-competition 1b). **May 2023**
- Selected for the [MITACS Globalink Research Internship](#). **Feb 2023**
- Selected for the [Visiting Student Research Program](#) at KAUST. **Oct 2022**
- Full-tuition scholarship funded by Corporación Gilberto Echeverri Mejía. **Dec 2017**
- Competitive programming & mathematics: 13th/36 ACM-ICPC South America/North Regionals and 11th/500+ RIMAC Hackathon (2022); 3rd/65 Ruta N coding competition (2020); 4th XII National Integration Bee (2019).

RESEARCH INTERESTS

Stochastic interpolants, information-theoretic estimation, score-based generative models, Bayesian neural networks and uncertainty quantification.